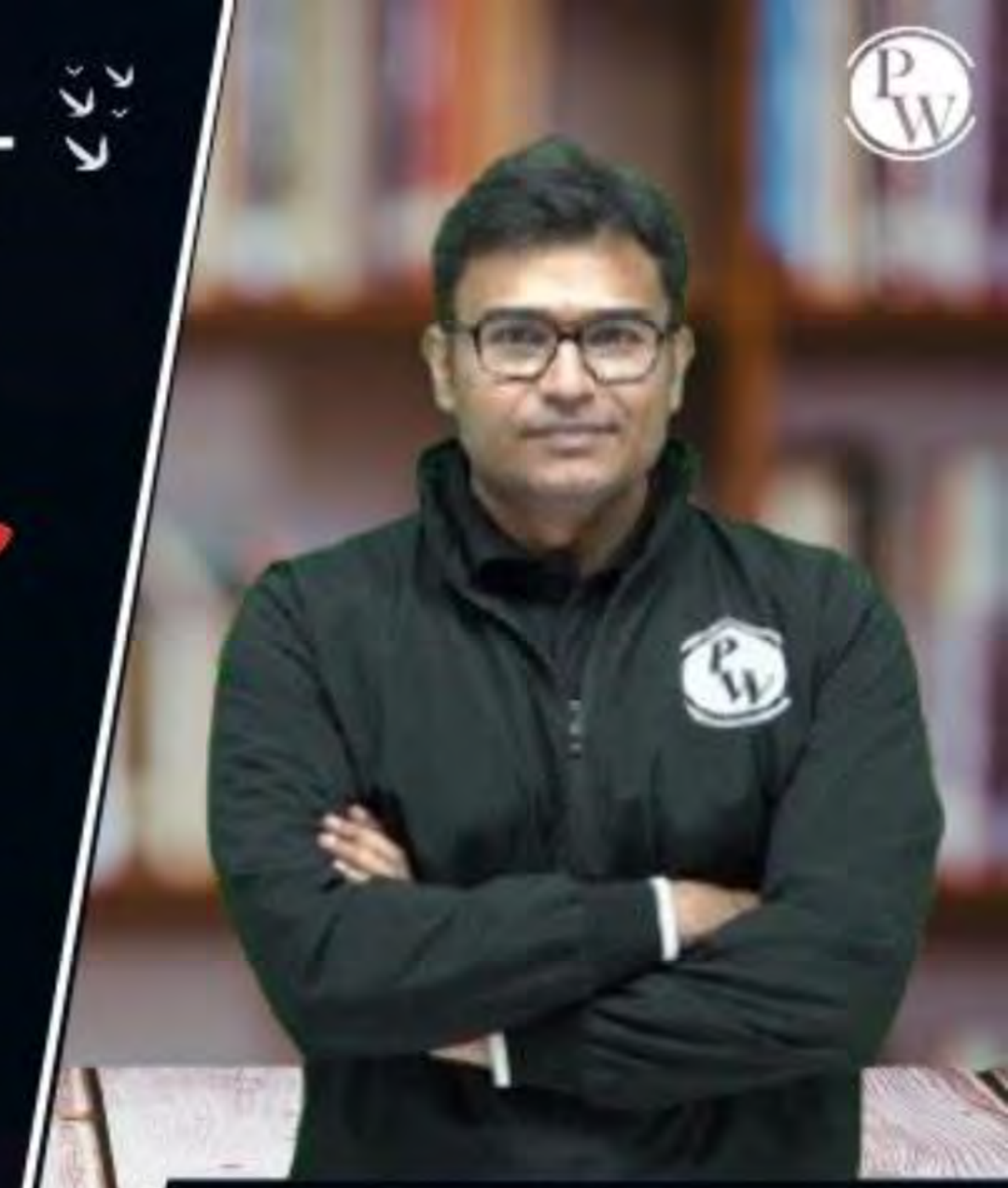
Computer Science & IT

Theory of Computation



Regular languages

Lecture No. 5



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

DFA, NFA, RE

Topics to be Covered



- Topic
- Topic
- Topic

DFA, NFA & RE

Properties of RE

How to Score full marks in TOC

1) Regular languages < Questions
   - Ullman
   - Peterliz
   - Problems
     - Tables
2) PDA
3) TM

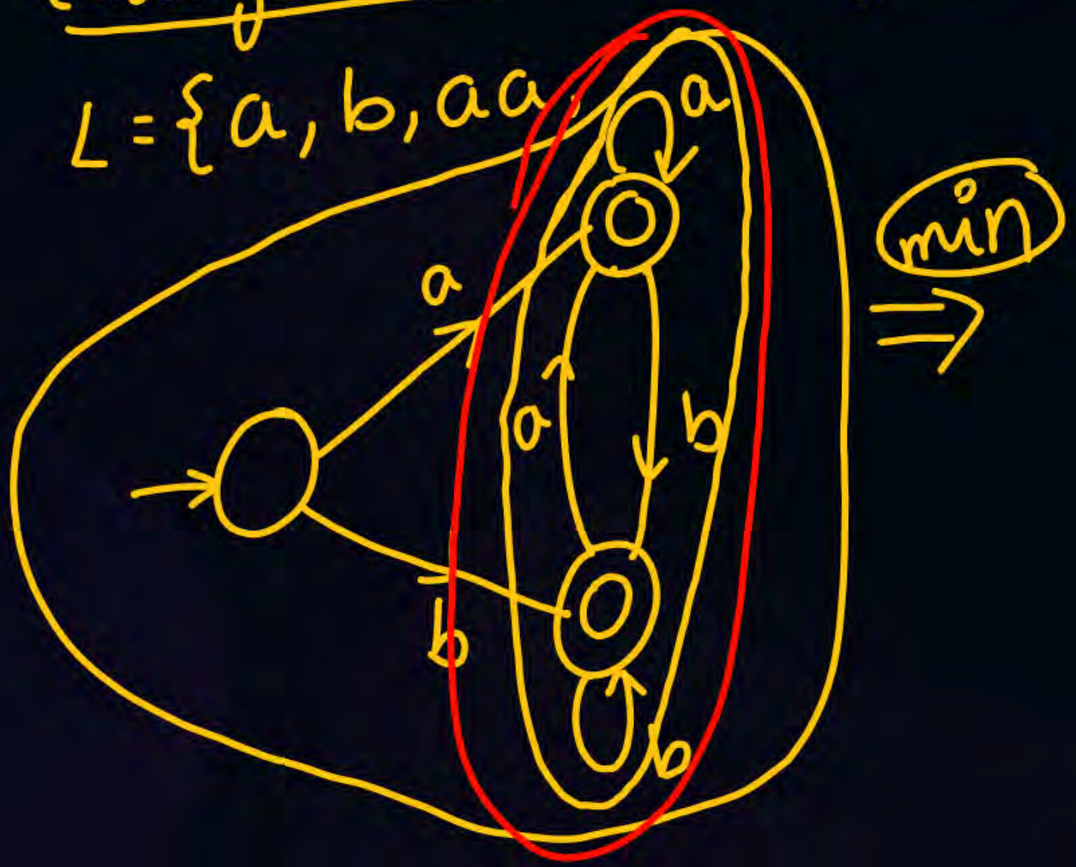


$(a+b)^*(a+b)$

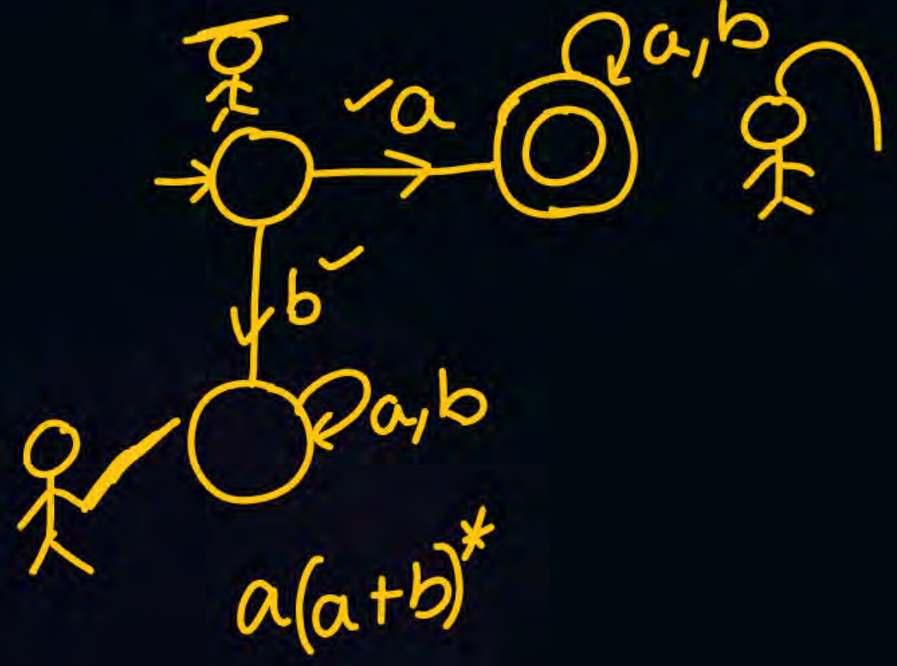
$(a+b)^*a + (a+b)^*b$

ending with a & ending with b

$L = \{a, b, aa, \ldots\}$



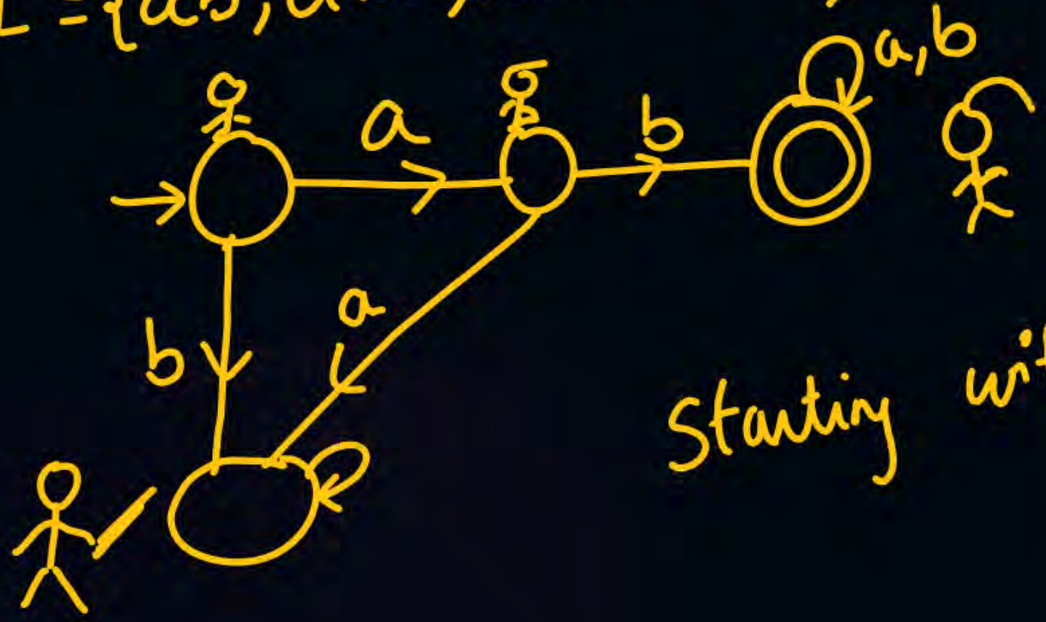
$\Sigma = \{a, b\}$ $L = \{ w / w \text{ starts with } a \}$



$a(a+b)^*$

$\Sigma = \{a, b\}$ $L = \{w / w \text{ starts with } ab\}$ $ab(a+b)^*$

$L = \{ab, aba, abb \cdots --\}$



Starting with aababa



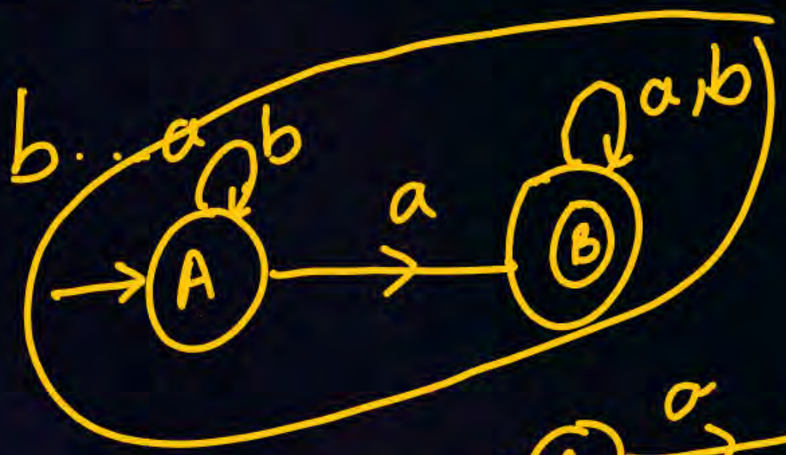
$\Sigma = \{a, b\}$ $L = \{w / w \text{ contains 'a'}\}$ ✓$(a+b)^* a (a+b)^*$

$b^* a (a+b)^*$ ✓

✓$(a+b)^* a\ b^*$

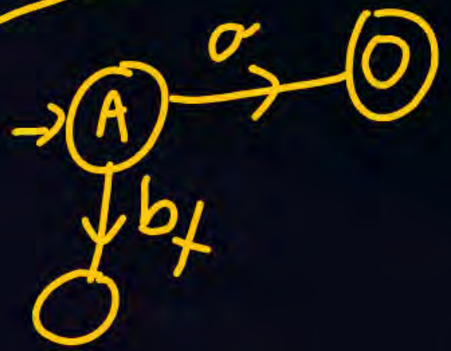


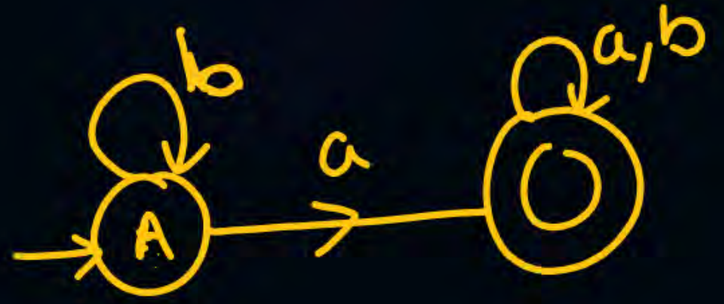
$L = \{a, ab, ba, abb, bab \ldots\}$

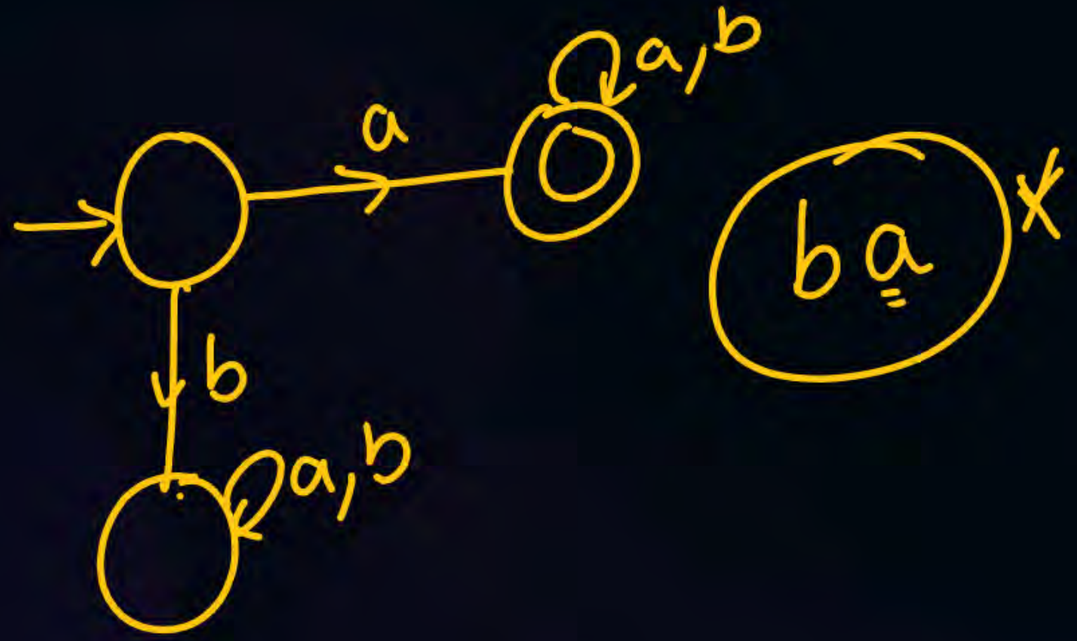


A → waiting for 'a'

B → I have seen 'a'



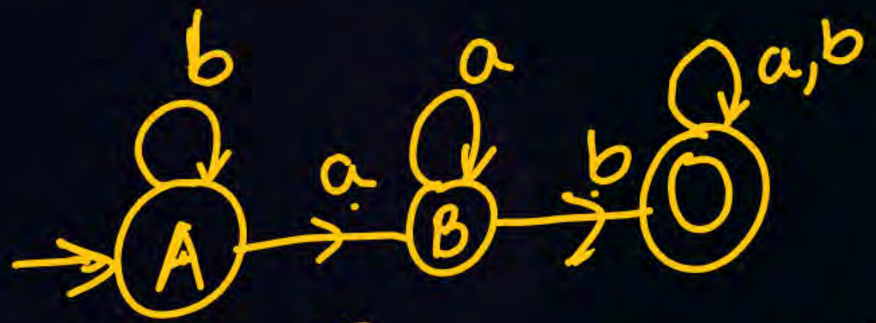

b
a
a,b
A


a
a,b
b
a,b
ba
X

$\Sigma = \{a, b\}$ $L = \{ w / w \text{ contain "ab" as a substring} \}$

$(a+b)^* ab (a+b)^*$ $L = \{ ab, aab, abb, aaab, \cdots \}$

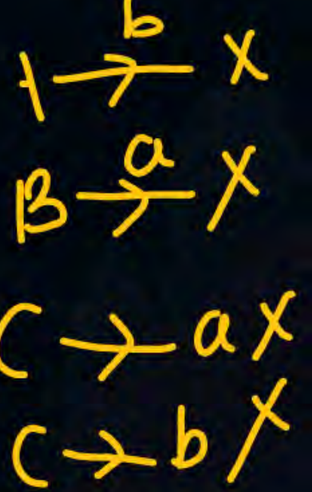




aaaab

aaaab

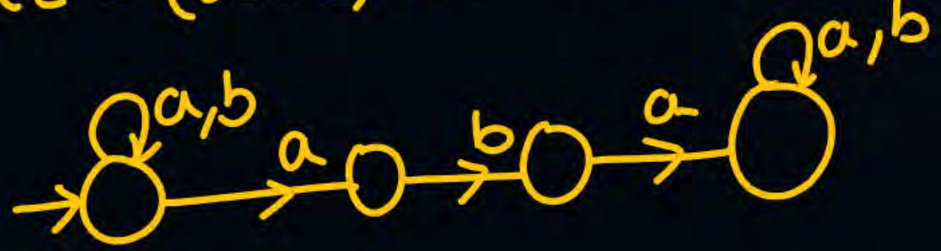
bbbab



$\Sigma = \{a, b\}$ $L = \{w / w$ Contains 'aba' as Substring $\}$

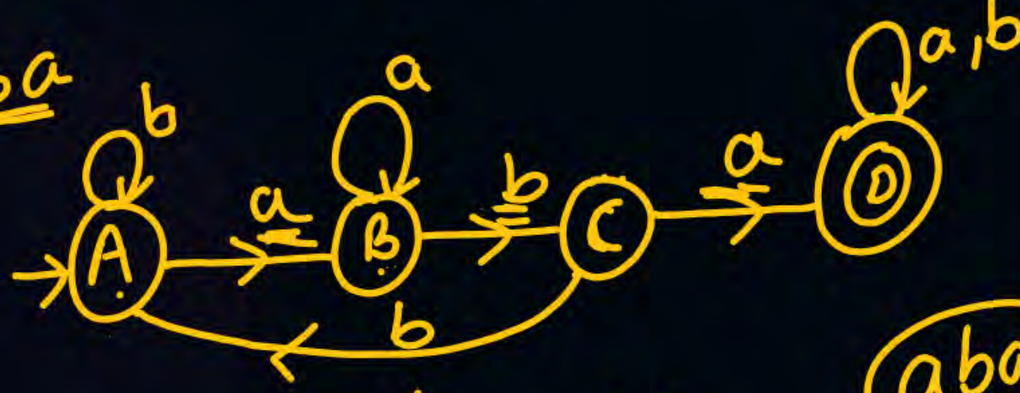
$RE = (a+b)^* aba (a+b)^*$

Starting with



$L = \{aba, a\underline{aba}, \underline{aba}a, b\underline{aba}, \ldots\ldots\}$

bbb<u>aba</u>





aa<u>aba</u>

aa<u>aba</u>

abbba

Starting with → We want something in the starting

Containing

_____( )

ending

$\Sigma = \{a, b\}$ $L = \{\omega / \omega \text{ ends with 'a'}\}$

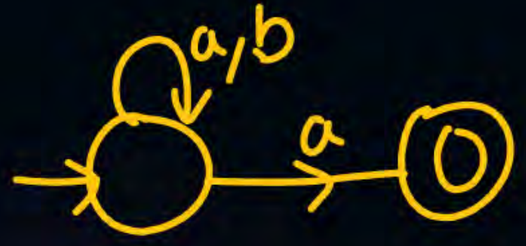
$RE = (a+b)^*a$

$(a+b)^*a$

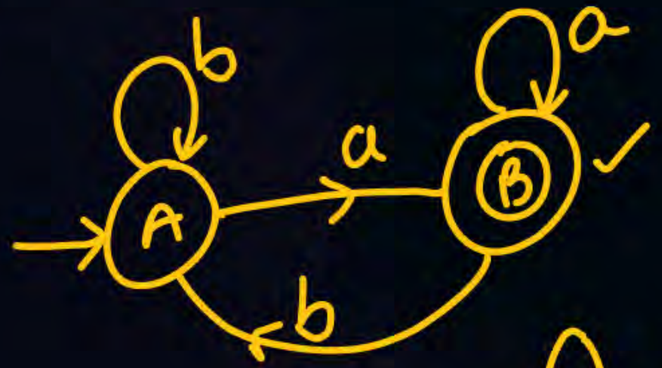
b^*a ✓

$L = \{a, aa, ba, \cdots\}$

NFA



DFA



A → waiting for last "a"

aaab ✓ bbba ✗

DFA → may Contain Dead State

min DFA → " " "

min NFA → No dead state

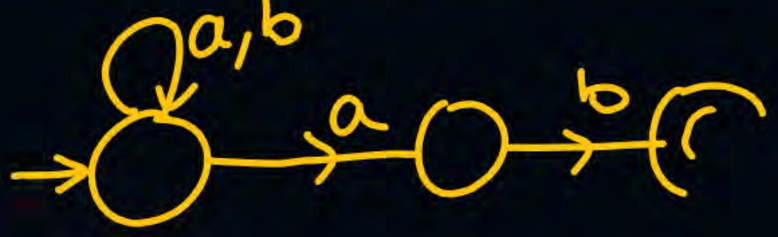
Starting with → dead state

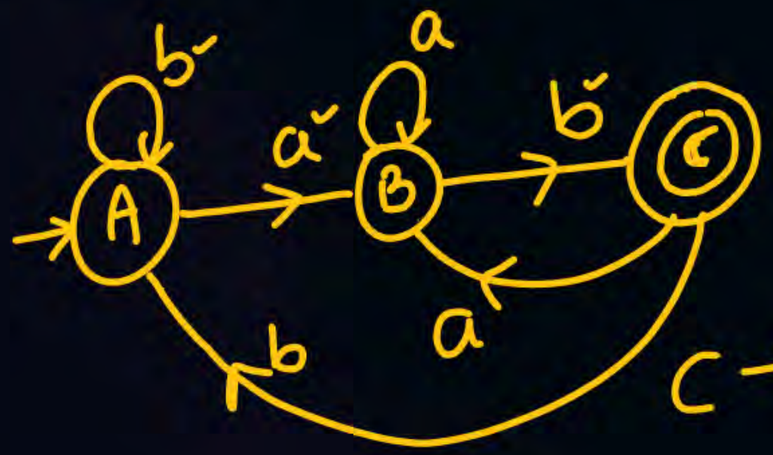
ending with and containing → No dead state

$\Sigma = \{a, b\}$ $L = \{w / w \text{ ends with } ab\}$

RE: $(a+b)^*ab$

$L = \{ab, aab, bab \ldots\}$





ab → 3

"abaaab" → 7

C → abb (ab)

Base 10 (1234) /10

10^2

10^3

10^4

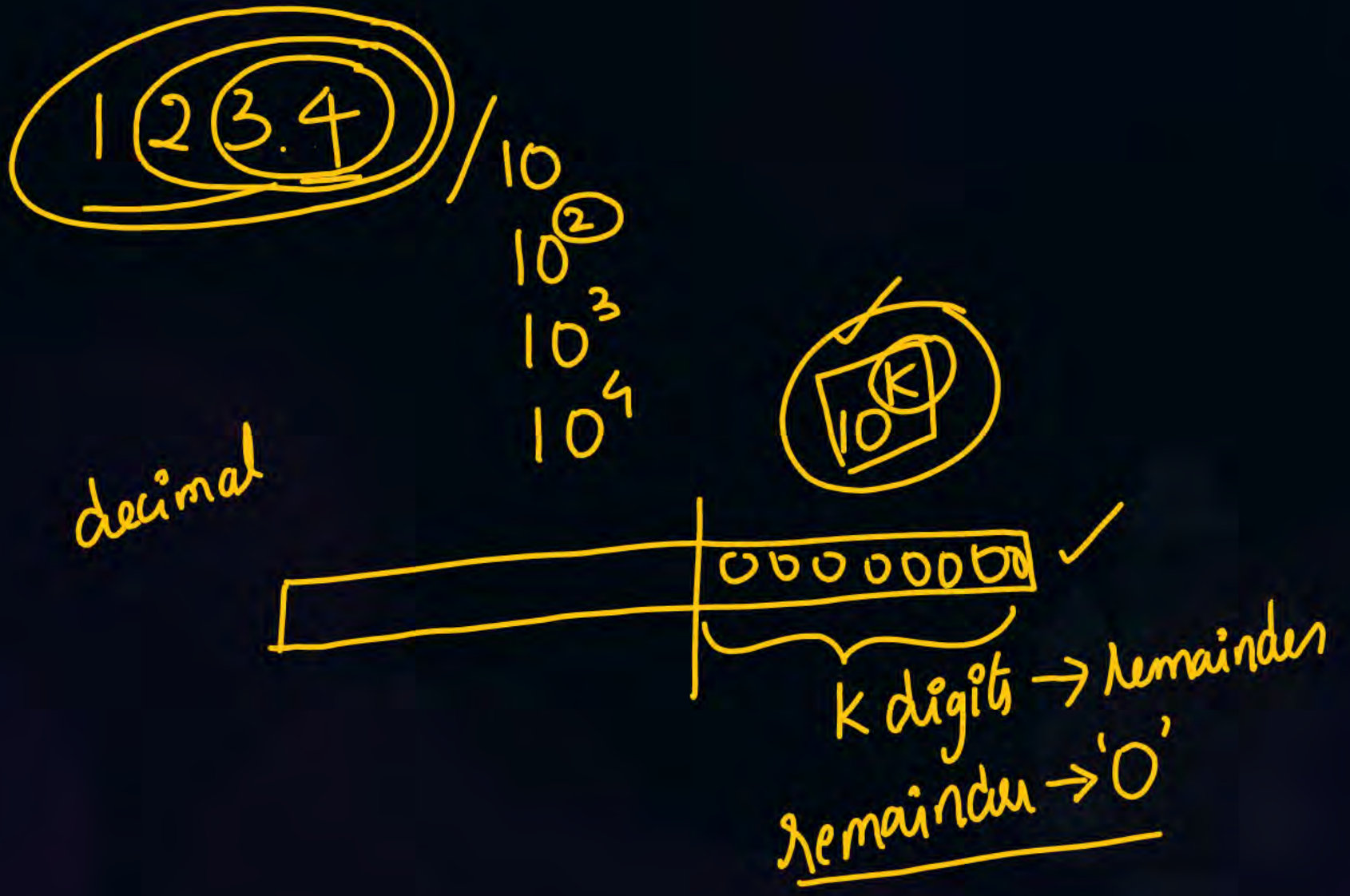
10^K

K digits → remainder

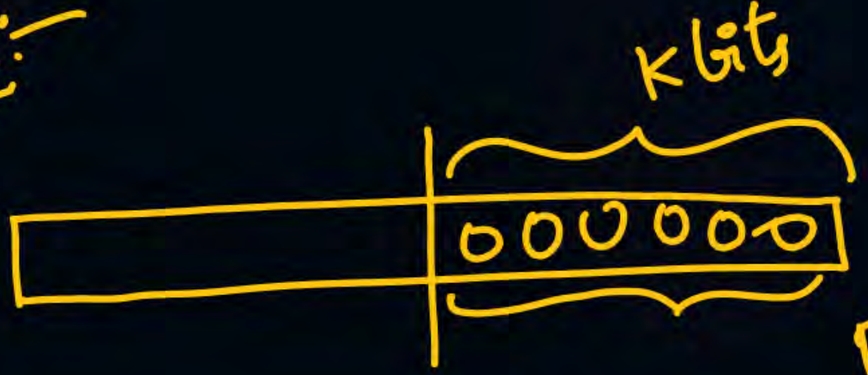
Base 10



10^K → remainder 0


1234
/10
10^2
10^3
10^4
decimal
10^K
K digits → remainder
remainder → '0'

Base 2 :-



Base 2

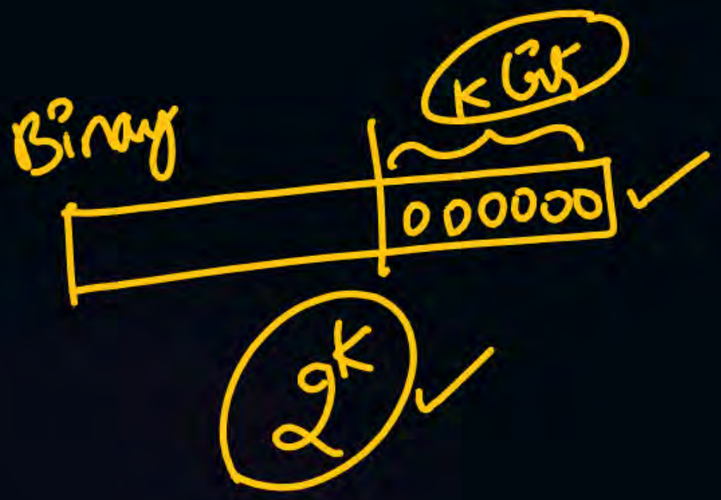
2^K ✓

Binary



2^k ✓

x is divisible by 2^K


Binary
K GK
000000
2K

$\Sigma = \{0, 1\}$ $L = \{\omega / \omega \text{ is divisible by } 2\}$

even



$(0+1)^*0$





$\Sigma = \{0, 1\}$ $L = \{\omega / \omega$ is divisible by $4\}$

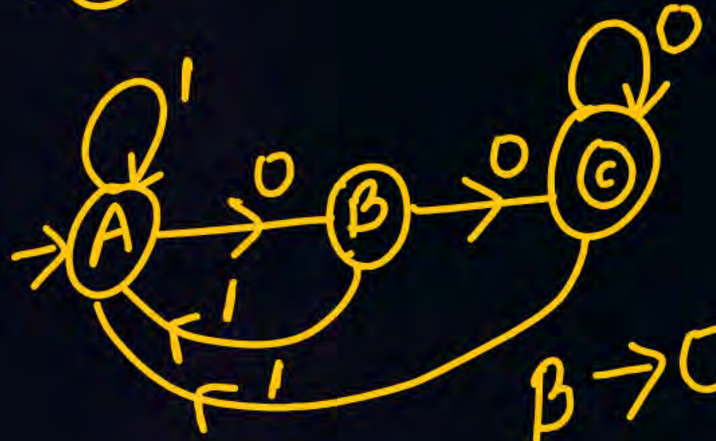


$2^{(2)}$

$L = \{00, 100, 000, \ldots\}$

$(0+1)^*00$





B → 0100

C → 001(∞)

$\Sigma = \{0, 1\}$ $L = \{w / w \text{ is divisible by } 8\}$

2^3 ending with 000

$(0+1)^*000$





$\Sigma = \{0,1\}$ $L = \{w / w$ is divisible by $2^k\}$

How many states in the min DFA ending with k 0's



$k \rightarrow (k+1)$ states

$\Sigma = \{0,1\}$ → Binary → $/2^1$, $/2^2$, $/2^3$ ..... $/2^k$

0, 00, 000

$\Sigma = \{0,1,2\}$ → Base 3 → $/3^1$ $/3^2$ $/3^3$ ..... $/3^k$

0 00 000

$\Sigma = \{0,1,2,3\}$ → Base 4 → $/4^1$ $/4^2$ $/4^3$ $/4^k$

0 00 000

k+1

$2^k, 3^k, 4^k$ ✗

$\Sigma = \{0, 1\}$ $L = \{ w / w \text{ is divisible } 3 \}$

THANK - YOU